# I · Monitor mount and accessories

The VIVO STAND-V300CU (P12) has two independent C-clamps and no base bar. Both clamp onto top A's own wall edges, never on the joint: clamp 1 centred at **x = 40.0 cm** on the back edge, clamp 2 centred at **y = 40.0 cm** on the side-wall end. The crossbar runs on the diagonal between the poles. Each lower jaw bears on a spare steel mending plate (P11), because particleboard crushes under a bare steel jaw over months. Datum: top A's back-left corner (0, 0); x along the back wall, y along the side wall.

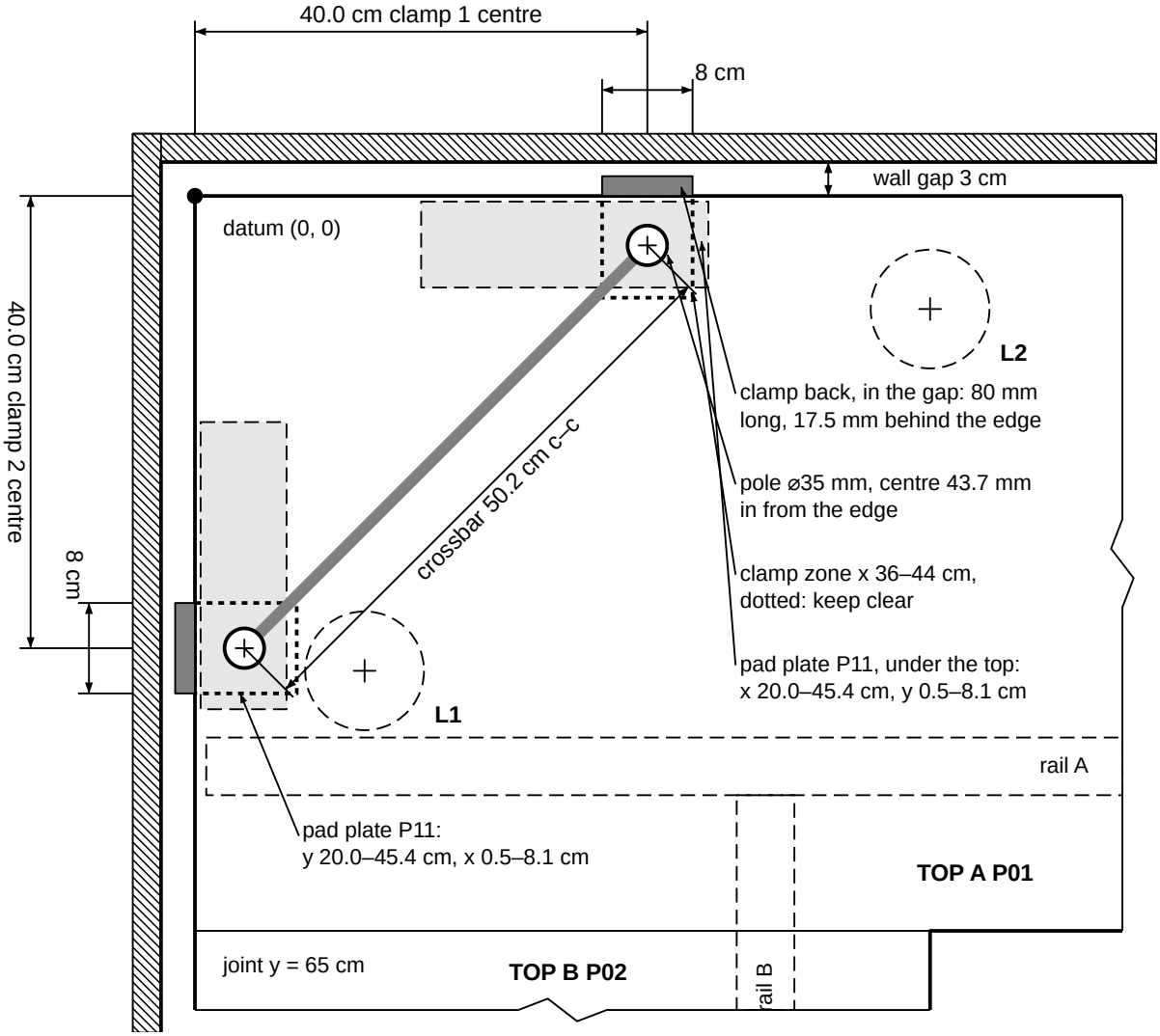



**Fig. I-1** Plan of the corner from above; tape readings in cm, caliper sizes marked mm. Dashed: under the top (pad plates P11, legs L1 and L2, rails). Dotted: the clamp zones, 36–44 cm along the edge, keep clear. Dark: the clamp backs in the 3 cm wall gap. Black circles: poles, centre 43.7 mm in from the edge.

| Balloon | Part | Balloon | Part |
|---|---|---|---|
| P01 | Top A, EKBACKEN 249 × 65 cm, 28 mm thick | P11 | Mending plate 10 × 3 × 0.12 in, two used here as clamp pads |
| P02 | Top B, EKBACKEN 249 × 65 cm, 28 mm thick | P12 | VIVO STAND-V300CU (clamps, poles, crossbar, arms) |

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## What is measured at the clamp

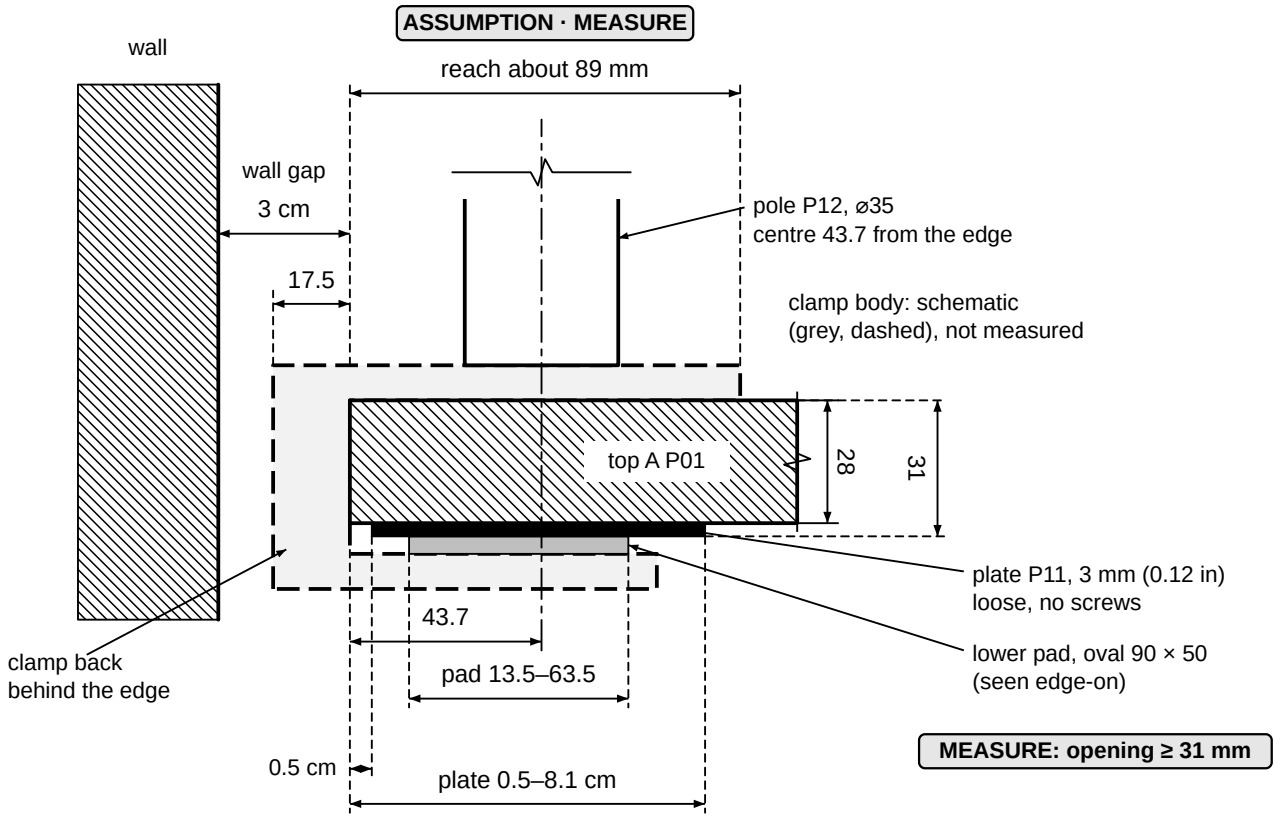



**Fig. I-2** Section through clamp 1 at x = 40.0 cm, looking along the back wall; mm unless marked cm. The clamp body is drawn as a plain C (grey, dashed) because its shape is not measured; only the dimensioned values are. The section through clamp 2 at y = 40.0 cm is the same, turned 90°.

| Value | Size | Confidence |
|---|---|---|
| Crossbar, centre to centre of its end collars (19¾ in) | 50.2 cm | VERIFIED |
| Pole diameter; pole centre in from the desk edge | 35 mm; 43.7 mm | VERIFIED |
| Clamp back: length along the edge; projection behind the edge (it sits in the 3 cm wall gap) | 80 mm; 17.5 mm | VERIFIED |
| Lower pad: oval 90 × 50 mm, 13.5 to 63.5 mm in from the edge, so it lands on the plate (0.5–8.1 cm from the edge) | 13.5–63.5 mm | VERIFIED |
| Clamp centre from the corner: ideal 39.9 cm; built at 40.0 cm, the crossbar sets the last millimetre | 40.0 cm | SOURCE-DEFINED |
| Clamp zones x 36–44 cm, y 0–9 cm (clamp 1) and y 36–44 cm, x 0–9 cm (clamp 2); pad plates x 20.0–45.4 cm, y 0.5–8.1 cm and y 20.0–45.4 cm, x 0.5–8.1 cm | Fig. I-1 | SOURCE-DEFINED |
| Top 28 mm plus plate 3 mm (0.12 in): under 60 mm, so VIVO's Option 1 holes (2 and 4 on the lower clamp) | 31 mm | SOURCE-DEFINED |
| Top arm reach from the edge | about 89 mm | ASSUMPTION / FIELD CHECK |
| Maximum jaw opening (must be 31 mm or more) | ≥ 31 mm | MEASURE ON HARDWARE |

## Hang the mount

1 **Lay the pad plates.** Lay a spare mending plate P11 flat on the underside, on each pad outline drawn when top A was marked, long side along the edge: x 20.0–45.4 cm, y 0.5–8.1 cm at clamp 1; y 20.0–45.4 cm, x 0.5–8.1 cm at clamp 2. Loose, no screws: the jaw holds it. SOURCE-DEFINED

> ! **MEASURE ON HARDWARE before hanging the clamps**
> Top arm reach (about 89 mm) and maximum opening (needs 31 mm or more: 28 mm top + 3 mm plate) are not measured. Open the lower clamp fully on the Option 1 holes and measure the gap with calipers; if it is under 31 mm, stop. The clamp zones are drawn 9 cm deep from the edge; if the reach measures more, keep the zone clear to the measured reach.

2 **Hang both lower clamps, finger-tight only.** Clamp 1 centred at x = 40.0 cm on A's back edge, clamp 2 centred at y = 40.0 cm on A's side-wall end (Fig. I-1). Use the VIVO sheet's Option 1 holes (2 and 4): top and plate together are 31 mm, under 60 mm. The clamp backs sit in the 3 cm wall gap; the lower pad lands on the plate, 13.5–63.5 mm from the edge (Fig. I-2).

3 **Fit the poles and the crossbar as the spacing guide** (VIVO sheet step 3). The crossbar measured 50.2 cm centre to centre, so it fixes the pole spacing. **Only now tighten both clamps**, with the crossbar on. VERIFIED

4 **Refit the crossbar at monitor height** (VIVO sheet step 6): 40–43 cm above the top (123.0–126.0 cm from the floor).

5 **Hang the three screens, all landscape**, and set the heights in Fig. I-3. The G5 goes on the centre bracket (VESA 75), screen centred on the diagonal at (36.5, 36.5) cm, 20 cm in front of the crossbar. The Acers go on the side arms, toed in 15°; each Acer's near end ends up about 12 cm from the G5's curved end. The side arms run along the wall behind each panel, then forward into the VESA plate.

### Weights against the mount's ratings

| Screen | On the mount | Rating | Confidence |
|---|---|---|---|
| Samsung Odyssey G5 32 in P13 (1000R, VESA 75), centre | about 5 kg | centre bracket 48.4 lb curved (59.4 lb flat); top edge at most 26.5 in (67.3 cm) above the top | SOURCE-DEFINED |
| Acer XZ320QU 31.5 in P14 (1500R, VESA 100), each side | about 5 kg (5.6 kg with its stand) | 17.6 lb (8 kg) per side arm | SOURCE-DEFINED |

> The clamps load top A's edges near legs L1 and L2, not the corner tip. Do not add a soft pad under a jaw: the steel plate is the pad.

## Heights above the top

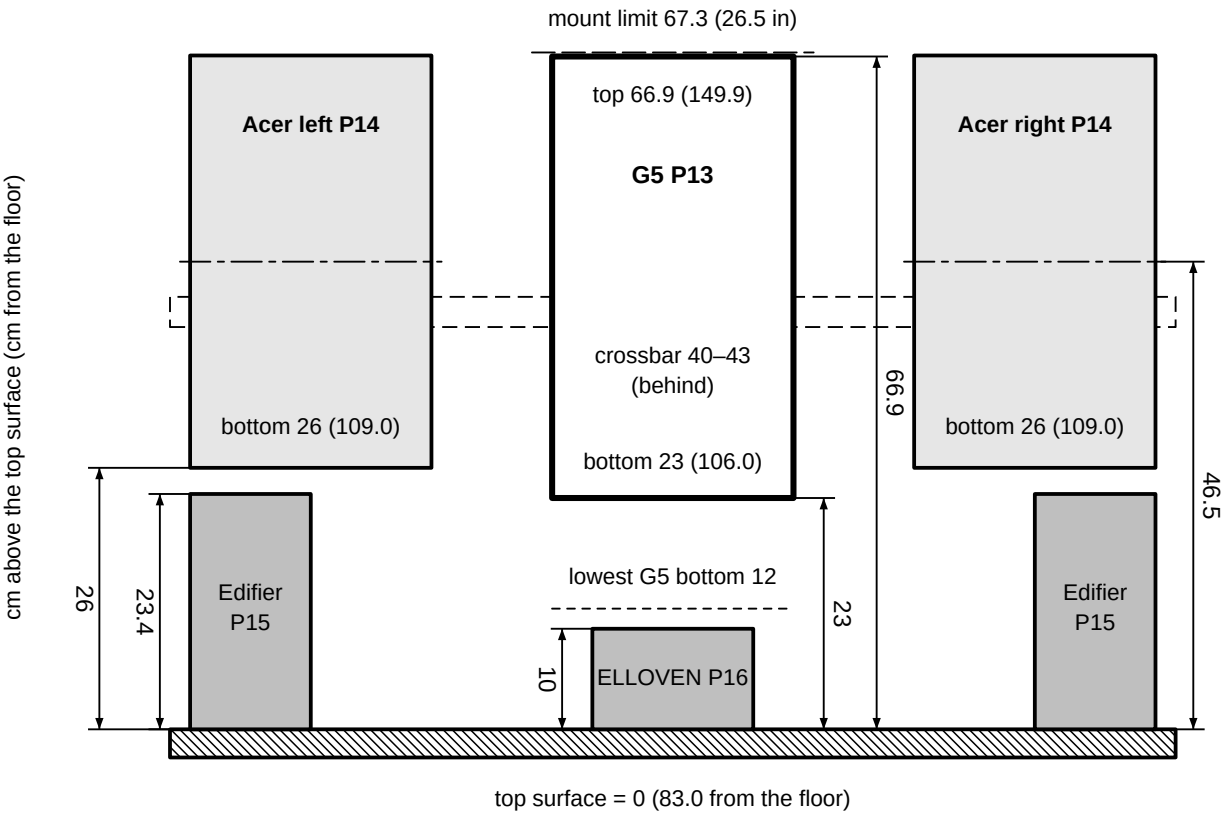



**Fig. I-3** The three screens seen from the seat, unfolded flat. Heights to scale, in cm above the top surface (in brackets, cm from the floor); widths are schematic. The Edifiers stand under the outer half of each Acer.

| Item (all cm) | Above the top | From the floor | Note |
|---|---|---|---|
| Top surface | 0 | 83.0 | underside 80.2, top 28 mm thick |
| ELLOVEN P16 top | 10 | 93.0 | 13 under the G5 |
| G5 bottom edge | 23 | 106.0 | SOURCE-DEFINED |
| Edifier P15 top | 23.4 | 106.4 | 2.6 under each Acer |
| Acer bottom edges | 26 | 109.0 | panel centre 46.5 above the top |
| VIVO crossbar | 40–43 | 123.0–126.0 | refitted at monitor height |
| G5 top edge | 66.9 | 149.9 | 0.4 under the mount's 67.3 (26.5 in) limit |

**IF THE ASSEMBLED MOUNT IS LOWER**
The G5's top edge at 66.9 cm is only 0.4 cm under the 67.3 cm limit. If the assembled mount will not reach, lower the G5: a bottom edge anywhere down to 12 cm above the top still clears the ELLOVEN (10 cm tall).

## Everything else on the desk

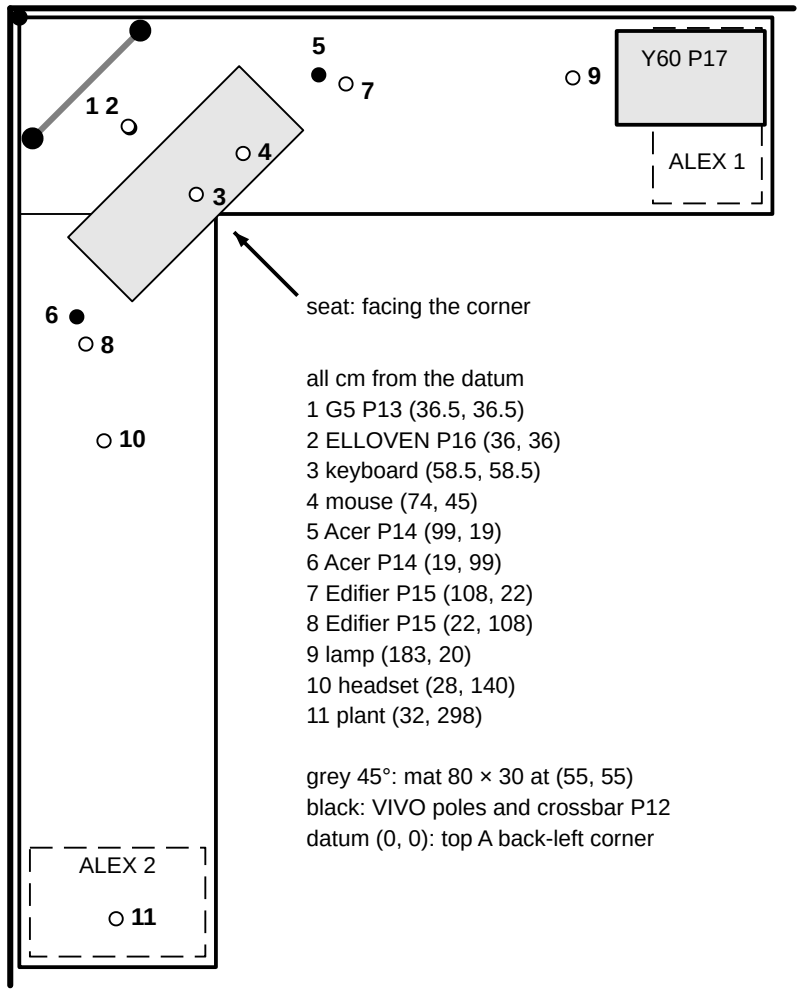



**Fig. I-4** Plan of the desk, centres in cm from the datum, top A's back-left corner. Dashed: ALEX 1 and ALEX 2.

ELLOVEN at (36, 36) cm, turned 45°, drawer toward you: its back corners clear the clamp plates and poles by over 2 cm. At (33, 33) cm it would sit on the clamps.

The mat's back edge touches the ELLOVEN's front feet; its front edge reaches the inside corner of the L.

The Y60 (P17) stands on its I/O end on the printed pedestal over ALEX 1: long axis along the back wall, side glass toward you, front glass up. Keep it there, not mid-wing.

| Item | Centre (x, y), cm | How |
|---|---|---|
| G5 P13 | (36.5, 36.5) | on the diagonal, 20 cm in front of the crossbar |
| ELLOVEN P16 | (36, 36) | turned 45°, drawer toward you |
| Deskmat 80 × 30 cm | (55, 55) | turned 45° |
| NuPhy Air75 V2; mouse | (58.5, 58.5); (74, 45) | keyboard turned 45°, on the mat |
| Acer P14, right; left | (99, 19); (19, 99) | along each wall, toed 15° toward you |
| Edifier P15 ×2 | (108, 22); (22, 108) | grilles toward you, toed with the Acers |
| HYTE Y60 P17 | x 197.5–246.5, y 4.5–35.5 | on its pedestal, over ALEX 1 |
| ELYONA lamp | (183, 20) | beside the Y60, arm toward the keyboard |
| Headset stand; plant | (28, 140); (32, 298) | on top B, hook toward you; plant over ALEX 2 |

All positions **SOURCE-DEFINED** (docs/build-manual.md step 8).

# J · Final inspection and load testing

Ten checks, in the build manual's order, before you call the desk done. Each is an ACCEPTANCE CHECK: a pass or fail you can see or measure. Several of them test a STRUCTURAL ESTIMATE from the design review that has not been measured on this desk; the check says which. Write the results in the boxes and in the sign-off table at the end of this chapter. Datum for every drawing: top A's back-left corner (0, 0).

## Check 1 · Surface height at the six corners ACCEPTANCE CHECK

**1**

Measure from the floor to the top face at each of the six corners of the L (Fig. J-1). Every reading **83.0 cm ± 0.3 cm** (82.7 to 83.3 cm). With the optional ALEX feet fitted, read instead: 2 cm feet 85.0 cm, 4 cm feet 87.0 cm, 6 cm feet 89.0 cm, each ± 0.3 cm. If a corner is out, turn that corner's OLOV foot or round foot, then measure all six again.

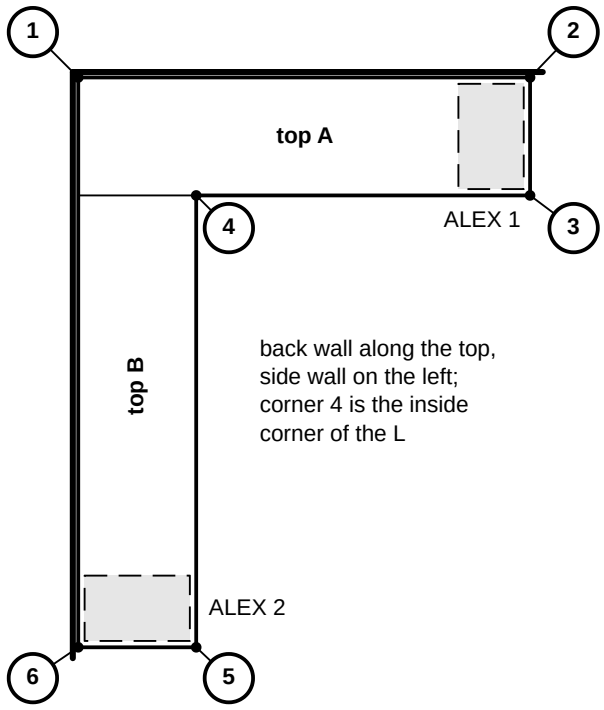



**Fig. J-1a** The six corners, numbered for the record.

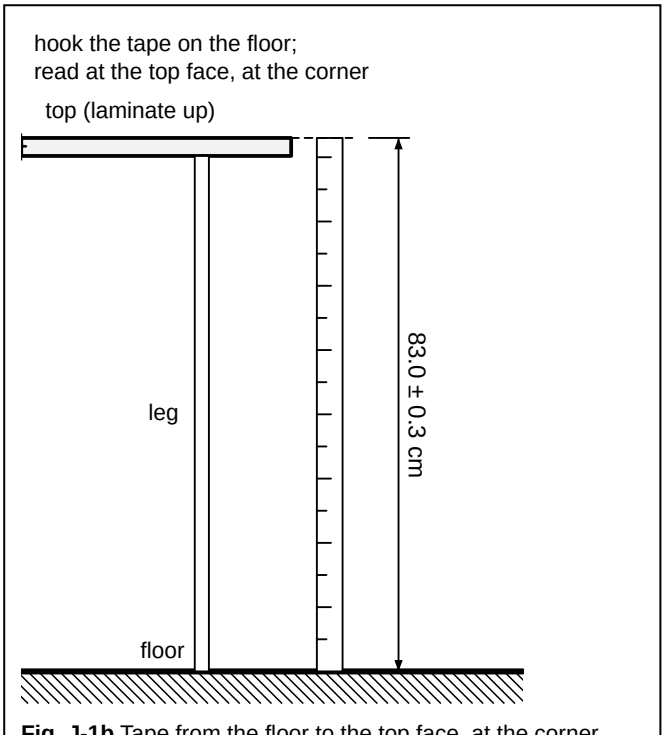



**Fig. J-1b** Tape from the floor to the top face, at the corner.

| Corner | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Reading (cm) | | | | | | |

## Check 2 · Level and flush across the joint ACCEPTANCE CHECK

**2**

A level across the joint reads flat both ways: across it (1) and along it (2). A fingertip drawn across the joint (3) finds no step. Level the two tops with the OLOV feet and the round feet, never by packing anything under the joining plates.

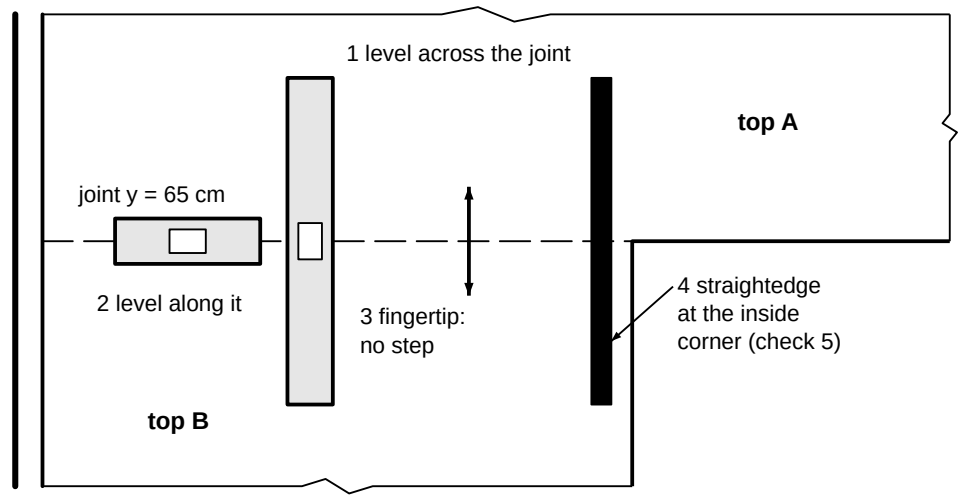



**Fig. J-2** Plan of the joint at y = 65 cm. 1, 2: level positions; 3: fingertip; 4: where the straightedge of check 5 goes, at the inside corner.

## For the remaining checks you need

**Tape**
floor to top face, checks 1 and 8

**Level, straightedge**
checks 2, 3 and 5; a rule to read the gap

**20 kg**
a case of water plus a few books, check 3

**Wrench**
torque wrench at 25 N·m, or a 15 cm wrench, checks 5 and 10

Do the checks in this order, with the screens and everything else in place.

## Check 3 · Corner load test ACCEPTANCE CHECK

> **NEVER STAND OR SIT ON THE DESK**
> The test load is a known weight: 20 kg of water and books. Do not stand or sit on the corner; that is not what the beam was sized for.

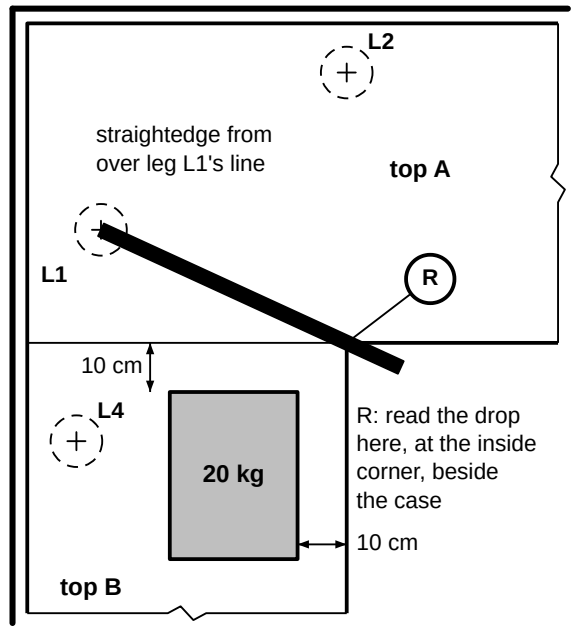



**Fig. J-3a** Plan. Case faces 10 cm in from top B's front edge and from the joint (case size schematic). R: reading point.

**3**

Set **20 kg** (a case of water plus a few books) on top B's inside corner, **10 cm in from both edges**.

Rest a straightedge from leg L1's line across the corner. Read the drop at the corner (R): **no more than 2 mm**. The estimate is about 1 mm. STRUCTURAL ESTIMATE

Lift the weight off. The corner returns to where it was, nothing creaks, no screw or bolt has moved.

Tests: sag at the inside corner (rail A carrying rail B's corner reaction) and corner-joint slip (the six fitting bolts holding by friction). STRUCTURAL ESTIMATE

| Drop under 20 kg (mm) | Returned? | Creak / movement? |
|---|---|---|
| | | |

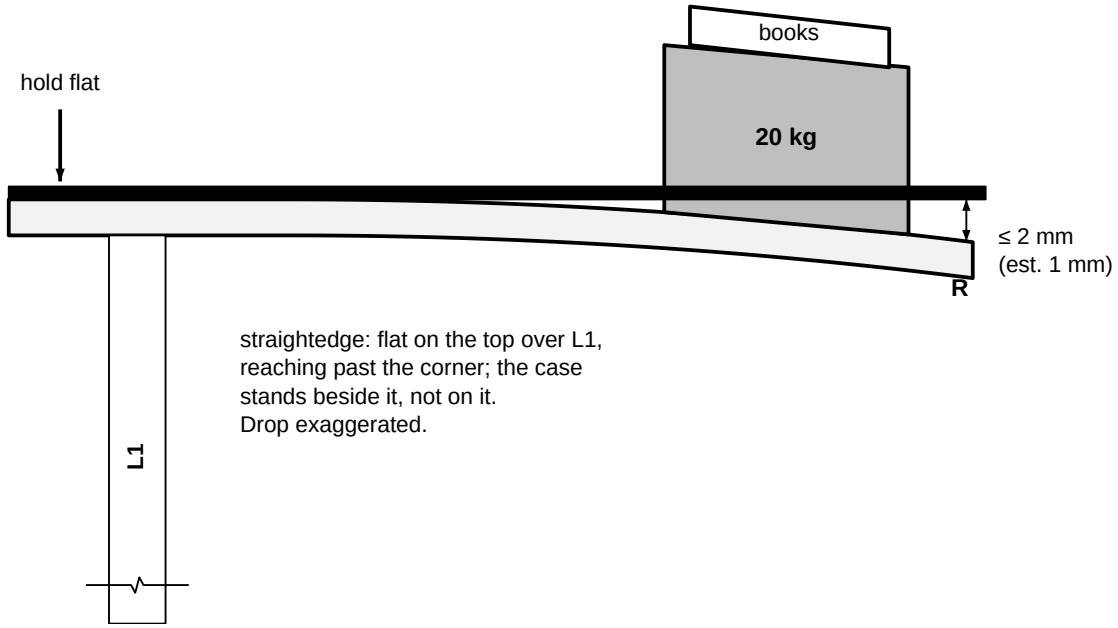



**Fig. J-3b** Elevation along the straightedge, from leg L1 to the corner. The case stands beside the straightedge, not under it.

## Check 4 · All eight pads bear ACCEPTANCE CHECK

**4**

At each of the eight feet F1–F8, try to turn the pad in its ring by hand. None spins free or rocks. A pad that spins or rocks is not bearing: see chapter K, "Top moves on its ALEX".

Tests: the racking estimate (eight pads captive in their rings hold the frame against a sideways push) and the ALEX load share. STRUCTURAL ESTIMATE

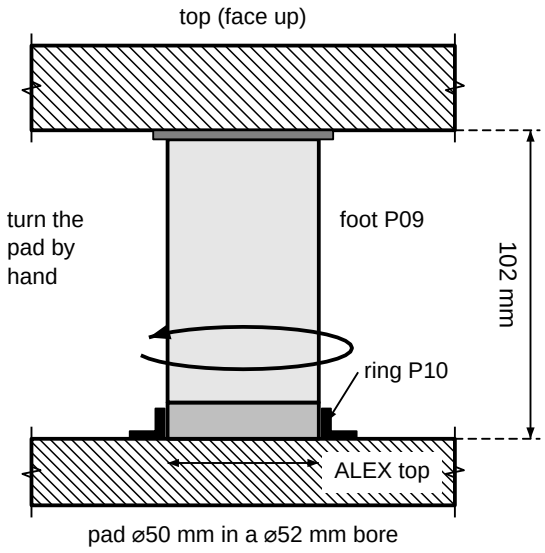



**Fig. J-4** Foot P09 on its pad in ring P10 on the ALEX top.

## Check 5 · Straightedge at the joint, now and later ACCEPTANCE CHECK

**5**

Lay a straightedge across the joint at the inside corner (Fig. J-2, position 4) and **write down any step**. Check it again after two weeks and after the first month. If the corner has settled, retighten the six fitting bolts (25 N·m, chapter K).

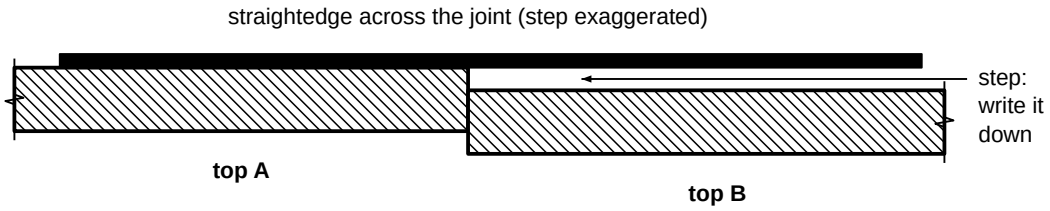



**Fig. J-5** Section across the joint at the inside corner. Step drawn exaggerated.

| Step at the joint | Day 0 | Two weeks | One month | Bolts retightened? |
|---|---|---|---|---|
| mm, and which top is higher | | | | |

Tests: corner-joint slip, an estimate, not a measured capacity. STRUCTURAL ESTIMATE

## Check 6 · Sideways push ACCEPTANCE CHECK

6

Push the desk sideways along each wall. No sway at the top beyond a few millimetres, and no top moves on its ALEX.

Tests: racking. The OLOV legs give almost no racking resistance; the two ALEX units, locked to the tops by the pads in their rings, do. STRUCTURAL ESTIMATE

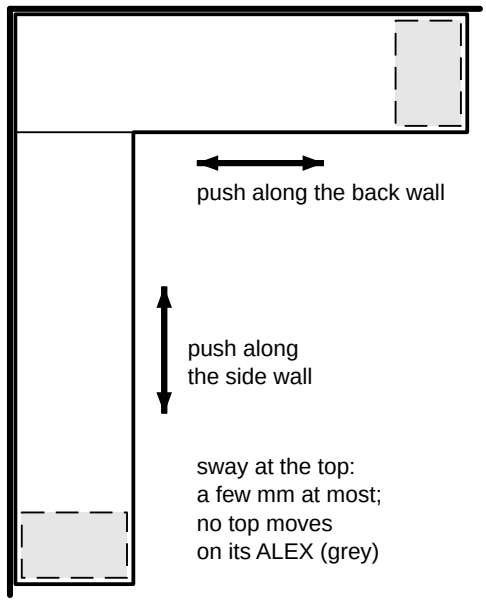



**Fig. J-6** Push along the back wall, then along the side wall.

## Check 7 · Screw tips ACCEPTANCE CHECK

7

**No screw tip** visible, or felt as a bump, on either laminate face. Run a hand over both tops, above every leg, foot plate, rail screw and joining plate.

## Check 8 · Knees clear ACCEPTANCE CHECK

8

**Knees clear.** Sit at the corner and slide in: nothing within reach of a knee or thigh. The rails' lower edge is 75.1 cm from the floor, 12 cm back from the front edge.

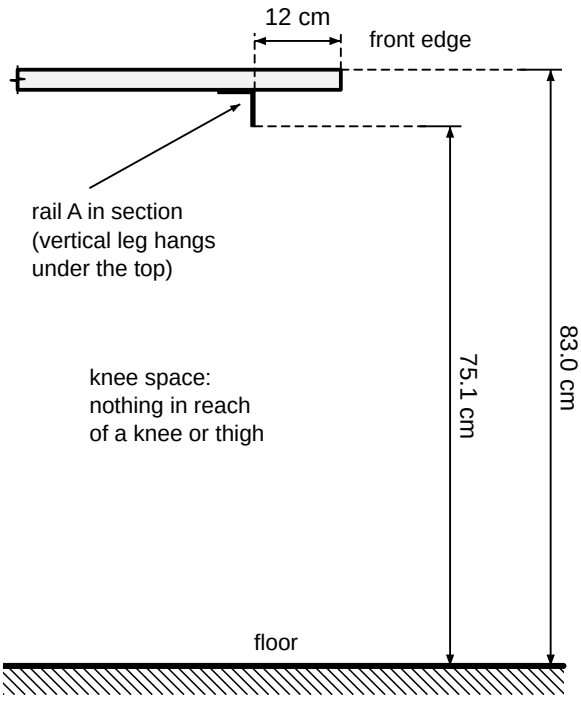



**Fig. J-7** Section through top A's front at rail A.

## Check 9 · ALEX fronts ACCEPTANCE CHECK

9

**Both ALEX fronts open fully** (ALEX 1's drawers, ALEX 2's door) without touching a foot or a leg.

## Check 10 · Retighten ACCEPTANCE CHECK

10

**Retighten every screw after two weeks**, then repeat check 5.

## Sign-off

| # | Acceptance check | Pass when | Structural estimate it tests | Result |
|---|---|---|---|---|
| 1 | Surface height, six corners | 83.0 cm ± 0.3 cm each (85.0 / 87.0 / 89.0 cm with 2 / 4 / 6 cm ALEX feet) | — | |
| 2 | Level and fingertip across the joint | flat both ways; no step felt | — | |
| 3 | Corner load test, 20 kg | drop ≤ 2 mm; returns; no creak; nothing moved | sag at the inside corner (estimate about 1 mm); corner joint slip | |
| 4 | All eight pads bear | none spins free or rocks | racking; ALEX load share | |
| 5 | Straightedge at the joint | step written down; re-checked at two weeks and one month | corner joint slip (settling) | |
| 6 | Sideways push along each wall | sway a few mm at most; no top moves on its ALEX | racking | |
| 7 | Screw tips | none visible or felt on either laminate face | — | |
| 8 | Knees clear | nothing in reach; rail lower edge 75.1 cm from the floor, 12 cm back | — | |
| 9 | ALEX fronts | open fully, touch no foot or leg | — | |
| 10 | Retighten every screw | done after two weeks | — | |

ACCEPTANCE CHECK pass or fail on this desk. STRUCTURAL ESTIMATE a number from the design review that the check confirms or refutes. If a check fails, see chapter K.

| Checked by | Date | Two-week re-check | One-month re-check |
|---|---|---|---|
| | | | |

# K · Maintenance and troubleshooting

## Schedule

| When | Do | See |
|---|---|---|
| **Day 0** | Write down the step at the joint (straightedge at the inside corner). | Check 5, Fig. J-5 |
| **Two weeks** | Retighten every screw. Straightedge at the joint again; if the corner has settled, retighten the six fitting bolts to 25 N·m. Look at each printed ring and its three screws. | Checks 5 and 10; troubleshooting |
| **One month** | Straightedge at the joint again; retighten the six fitting bolts if it settled. | Check 5 |
| **Any time** | If the corner ever drops a millimetre, retighten the six fitting bolts. | Corner bolts, below |

> ! **NEVER DEEPER THAN 21 MM INTO A TOP**
> The tops are 28 mm particleboard with a thin laminate skin; a screw reaching the skin lifts it. Replace a screw only with the same size. If a top measured 27 mm, shorten every screw into it by 1 mm (build manual step 0). SOURCE-DEFINED

## Corner bolts

**Bolts** six 3/8-16 × 1-1/2 in hex bolts, grade 5 (H08), washer under head and nut (H09), nylon-insert nuts (H10): four through rail A's vertical leg, two through rail B's leg with a fitting on each side

**Torque** 25 N·m (18 ft·lb); without a torque wrench, pull as hard as one hand can on a 15 cm wrench, then stop

**Why** a friction joint: each fitting leg has two 21 × 12 mm slots, centred 29.15 mm and 78.45 mm from the heel VERIFIED ; they leave each bolt 2.5 mm of play across and 11.5 mm along, and the rails' 7/16 in holes 1.6 mm, so only the clamping force keeps rail B's corner from settling

**Order** tighten only with the two surfaces flush and rail B's end touching rail A

The mending plates across the joint align the tops; they do not carry the corner. SOURCE-DEFINED

## Troubleshooting

| Symptom | Cause | Fix |
|---|---|---|
| **Corner has dropped; step at the joint grew** | The six fitting bolts lost clamping force (friction joint); supports settling | Bring the surfaces flush with the OLOV feet and the round feet, never by packing anything under the joining plates. Check rail B's end still touches rail A. Retighten all six bolts to 25 N·m. Write the new step down (Fig. J-5). |
| **Top moves on its ALEX**, or a pad spins or rocks | A pad is out of its ring or not bearing | Lift that ALEX end a few millimetres and slide the ALEX unit, not the top, until the pads drop into their rings. Each foot set with the 102 mm foot height gauge, no rock. Level along x and y; re-check 83.0 cm at the corners (Figs. J-1, J-4). |
| **Printed ring cracked** or its screws loose | Ring wall overloaded by a sideways push | Inspect at two weeks. A crack: reprint the ring (parts/structure/foot-ring.stl), PETG or ABS, flange-down: 52 mm bore, 3 mm wall, 75 × 2.5 mm flange. Refit with #6 × 5/8 in flat-head screws (H06), 5/64 in pilots 12 mm deep (12 ALEX puck). Never run the desk with a ring missing. |
| **Leg moves or wobbles** | OLOV plate screws loose | Retighten the five M4 × 16 mm screws (H01). A screw that spins: repair the hole (below), re-pilot 1/8 in, 15 mm deep (15 LEG puck). Re-set the OLOV to 80.2 cm, or the ALEX-feet table's number. |
| **Monitor clamp creeps** | Clamp loosened over time | Retighten the clamp with the crossbar fitted. Keep the steel plate (P11) flat on its outline under the jaw. Never add a soft pad under the jaw. |
| **Screw spins** (stripped) | Hole worn in the particleboard | Back it out, fill the hole with wood glue and a couple of matchsticks, let it dry, re-pilot. |
| **A mark is off by more than 2 mm** | Marking error | Redraw it. A stray 1/8 in hole in the underside does no harm. |
| **A fitting bolt will not go in** | Holes not quite in line | Open the hole in the rail with a round file; never force it. |

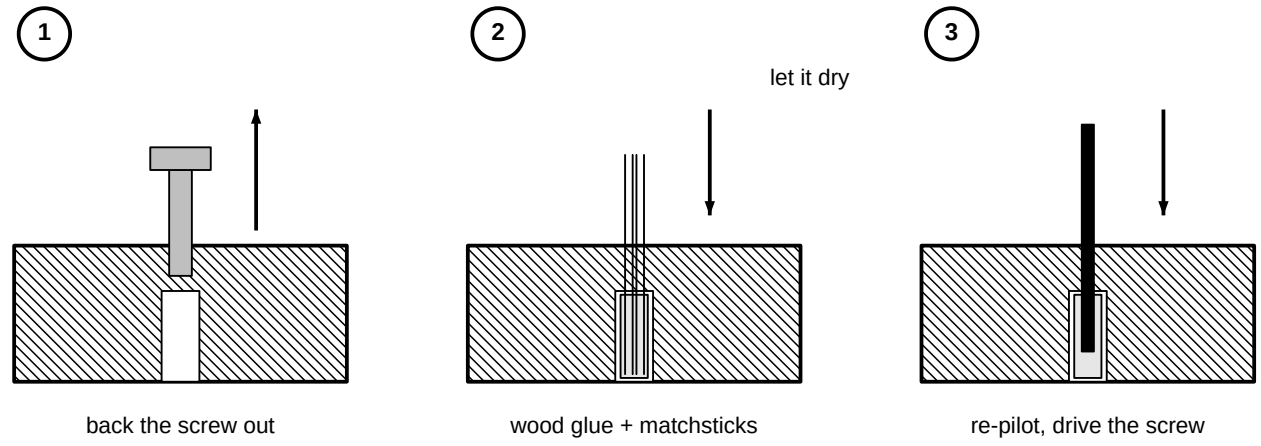



**Fig. K-1** A screw that spins: back it out, fill the hole with wood glue and a couple of matchsticks, let it dry, re-pilot with the step's own bit and depth.

Every fix above is from the build manual and the build guide SOURCE-DEFINED.

# Appendix · Outstanding measurements

These values were not measured on the purchased parts when this manual was built. Each is marked MEASURE ON HARDWARE at its point of use. Measure it before the step that uses it; if it differs, the step's own warning panel says what changes.

| Measurement | Expected | Used in |
|---|---|---|
| ALEX bottom panel thickness (only for the optional ALEX feet) | not yet known | Optional ALEX feet: three #6 × 5/8 in screws each. (The top panel is measured: 15.9 mm, VERIFIED.) |
| ALEX height, with and without its stock feet | 70.0 cm (docs assume) | Underside 80.2 cm; OLOV setting 80.2 cm; surface 83.0 cm (check 1) |
| Feet pad diameter (calipers) | 50 mm; rings bored 52 mm | Printed rings P10 before printing; check 4 |
| EKBACKEN thickness (calipers) | 28 mm | Every screw into a top (never deeper than 21 mm; if the top is 27 mm, shorten every screw by 1 mm); clamp opening 31 mm (chapter I) |
| VIVO top arm reach and maximum opening | reach about 89 mm; opening 31 mm or more | Chapter I: clamp zones, Fig. I-2 |
| Delivered rail lengths and angle thickness | 213 cm layout (bought as 84 in, 213.4 cm); 4.8 mm | Build manual step 1 hole positions; rail pilot depth 25 mm from the steel face |
| Room: wall to wall, and baseboard thickness | 252.0 cm and 317.0 cm | Build manual step 0 (252 × 317 cm of floor, including the 3 cm wall gaps); step 8b wall-gap trim |
| Feet load rating | unknown; the design wants 50 kg or more each | Load path through the eight feet; check 3 corner load test |

## Confidence labels

| | |
|---|---|
| VERIFIED | measured on the purchased part, or taken from its own geometry file. |
| SOURCE-DEFINED | defined by the project documents (docs/build-manual.md). A source-defined dimension is never called verified. |
| ASSUMPTION / FIELD CHECK | engineering estimate or field check. |
| MEASURE ON HARDWARE | not yet measured: measure it before the step that uses it. |

## Quality control

Every number in this manual comes from manual/data.json (extracted from guide.html's D object), manual/facts.json or docs/build-manual.md. Source: HEAD of https://github.com/scotthowson/desk-setup on 23 Oct 2026 (supersedes a2e99ea with measured facts). If you change a measured value, change it in the docs first, then rebuild the manual.